Assignment 2 — Tangible and Embodied UI for OpenGL Racing Game on a Table

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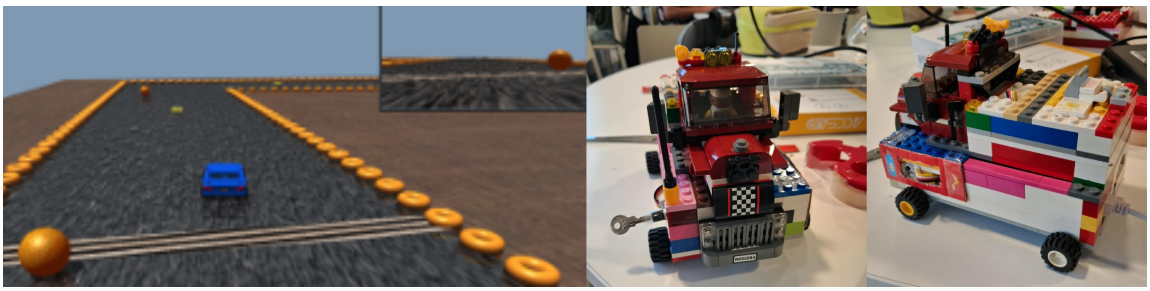


Fig. 1. Open GL Racing Game and the corresponding Tangible Interface

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Additional Key Words and Phrases: Car, Game, Interaction, Camera, Tangible

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1 Introduction

Our interface combines a 3D-printed car, which is both the representation of the game vehicle and the controller that drives it, with a camera that recognises physical objects shown by the players (coloured signs and a candle) and uses them to control the state of the game and its environment.

An Arduino inside the car reads an accelerometer, a potentiometer, a light sensor and a button, and drives an LED. Tilting the car diagonally accelerates and turns the virtual car, and holding it level stops it. Turning the potentiometer starts the engine, and the LED shows whether the engine is on or off. The button gives a boost. The light sensor simulates day and night by switching the global light of the game on and off.

The camera adds two commands that work around the car. A red sign held in front of the shoulder stops the game, and a blue sign held in the same position resumes it. A candle, when lit, turns on the candles inside the game.

2 Theoretical adherence

2.1 Pillar 1: Coupling and Phenomenology

Dourish (2001) [1] argues that interaction is a form of embodied action situated in the world, and that meaning arises through use rather than being fixed by the designer. Our interface follows this view in several ways.

First, it supports the phenomenological distinction between ready-to-hand and present-at-hand. During normal play the car should be ready-to-hand: the player thinks "I am taking the corner", not "I am tilting an accelerometer by twenty degrees". No menus or on-screen controls sit between intention and action, so the object disappears into the activity. When something breaks down, for example when the engine is off, the LED is dark and the car does not respond to tilting, the car becomes present-at-hand and the player's attention shifts to it.

Second, the coupling between the physical and virtual car is tight, continuous and reversible. Tilt maps to acceleration and direction in real time, the potentiometer switches the engine on and off, and the button adds a boost on demand. The player can turn the engine off, level the car or uncover the light sensor at any moment.

Third, the design follows Dourish's principle that users, not designers, manage coupling. Players decide when to start the engine, when to drive, when to boost and when it is night, and they choose how to hold the car: with one hand or two hands. The style of coupling is theirs.

Fourth, the light sensor follows the principle that embodied interaction turns action into meaning. Covering the sensor with a hand to "make it night" is a gesture that players can invent for themselves, and it carries meaning in the game world without any button label.

The camera channel extends the embodied-action argument. Dourish stresses that meaning is created and shared through action in a common space. Holding a red sign in front of the shoulder is a visible act that every player at the table can see and understand, much like a referee showing a card, so pausing and resuming the game becomes a social act and not a private command typed into a system. It relies on objects and body positions that players already know from daily life, which keeps it close to the ready-to-hand experience.

The candle shows a different kind of coupling, one that persists over time. The in-game candles stay lit while the physical candle burns and go out when the player blows it out, so the physical and virtual states remain in sync without any extra command. The player manages the coupling through the object itself and can undo it at any moment.

Finally, our game features a sports car, which raises the question of whether the Proteus Effect could be observed in this context. For example, could driving a sports car influence the user's perception of speed and acceleration? The car's

design and characteristics might influence the user's sense of power, confidence and driving behaviour, demonstrating how virtual representations can shape the user's perceptions and experiences.

2.2 Pillar 2: Material Semantics

Ullmer, Shaer, Mazalek and Hummels (Ch. 1-3) [4] describe tangible interaction as weaving digital information into physical form, so that the shape and material of an object carry meaning and give physical expression to digital state. Our interface applies this in five ways.

The form of the interface expresses its function. A car-shaped object is read as "the car" without instruction, which gives the interface a strong physical metaphor.

The components are placed where their meaning lives. The light sensor sits at the top of the car, receiving light directly from the "sky", which is why it controls the global light of the game. The potentiometer sits at the engine position. Each component's location tells the player what it does.

Material choices support the interaction. The car's weight and size allow it to be held and pushed like a toy.

The camera does not carry material meaning itself, so we place it in the background on purpose. The meaning lies in the objects it sees: the colour of the sign (red to stop, blue to resume) and the candle, whose flame is the very thing it represents in the game. This keeps the material semantics in the physical objects and prevents the camera from becoming a second, competing interface.

Digital and physical state remain linked. The LED makes the state of the engine perceivable on the object itself, and the candle makes the state of the in-game candles perceivable on the table, not only on a screen.

2.3 Pillar 3: Tangible Manipulation and Physical Affordances

Hornecker and Buur (2006) [2] organise tangible interaction around four themes, Tangible Manipulation, Spatial Interaction, Embodied Facilitation and Expressive Representation, and our design addresses all of them.

Tangible manipulation: Players grasp and move the real object (haptic direct manipulation). Small movements, either of the car or of a sign, give immediate feedback (lightweight interaction), and the mapping is easy to understand: when the car moves, the virtual car moves, and when someone shows the red sign, the game stops (isomorph effects).

Spatial interaction: Interaction uses the whole hand and arm around the car (Inhabited Space), and everything stays visible at once (Non-fragmented Visibility). The car's position, orientation and movement carry meaning (Configurable Materials, Performative Action). Holding a sign in front of the shoulder or lighting a candle are bodily actions performed in the shared space, and they change the state of the game (Full-Body Interaction).

Embodied facilitation: The physical form shapes behaviour, since the car can only be tilted or slid, the knob can only be turned, and the signs only count when they are held in front of the shoulder (Embodied Constraints). This position also separates a deliberate command from a sign that is merely visible in the scene. Several people can interact with the game at the same time using the car and the camera, for example one player drives while another holds the sign or lights the candle (Multiple Access Points). Using a car to control a car is how we built the experience of controlling the vehicle (Tailored Representation).

Expressive representation: The state of the engine is physically perceivable through the LED (Representational Significance). The signs let the user talk to the game, for example to tell it to stop or to resume (Externalisation). It is also easy to understand that the physical car and the virtual car represent the same object, and that the lit candle and the in-game candles represent the same flame (Perceived Coupling).

In terms of physical affordances, the knob affords turning, the button affords pressing, the car body affords grasping, tilting and sliding, and the signs afford being held up and shown.

2.4 Pillar 4: Intuitive Mapping and Image Schemas

Table 1. Johnson's account of embodied meaning [3] Interaction Mapping

Interaction	Image schema	Intuitive Mapping
Tilting the car diagonally accelerates and turns it that way	PATH, FORCE, BALANCE	An object tilted on a slope tends to move in the direction of the tilt
Holding the car level stops it	BALANCE	Level means stable and at rest
Turning the knob from zero starts the engine	ON-OFF, SCALE	Minimum means nothing is happening, and moving away from it activates the engine
LED lit means engine on, dark means engine off	ON-OFF, GOOD IS LIGHT	Light signals activity, dark signals rest
Pressing the button gives a boost	FORCE	A push produces a burst of extra force
Covering the light sensor makes it night	LIGHT-DARK, ON-OFF	Covering the "sky" darkens the world, as covering the sun does
Red sign in front of the shoulder stops the game	BLOCKAGE, FORCE	A raised sign acts as a barrier in the path of the game
Blue sign in front of the shoulder resumes the game	RESTRAINT REMOVAL, PATH	Removing the barrier lets the movement continue
Lighting the candle turns on the candles in the game	LIGHT-DARK, ON-OFF	A lit flame stands for itself, and light means active

3 Cyberphysical integration and execution

Regarding the cyber-physical integration and implementation, the initial idea was to place the Arduino and the sensors inside a 3D-printed car. By modelling dedicated spaces for each sensor and for the Arduino, it would be easier to organise and secure all the components, preventing them from becoming loose and potentially compromising the system's functionality. Furthermore, the 3D-printed material was expected to provide sufficient durability.

A 3D model was developed; however, it could not be printed in time for the peer critique. As a result, a paper prototype was used instead, which proved to be significantly more fragile and made it more difficult to arrange and secure the sensors (Figure 1). Since it was also not possible to produce the 3D-printed version for the final prototype, we had to adopt an alternative solution that would still ensure durability and allow for easy sensor organisation. We therefore decided to build the car using LEGO pieces.

The flexibility of LEGO components made it easier to position and secure the sensors in their appropriate locations. The final prototype is presented in Figure 2. The Arduino is positioned centrally within the car's base, as it is the heaviest component, helping to ensure a balanced weight distribution. The potentiometer is located on the right front, whereas the button is positioned on the left front, where the mechanisms usually are for the car functionalities. The LED is placed at the rear of the car, and the light sensor is located on the top surface facing the "sky".

The accelerometer is positioned close to the object's centre of gravity to ensure more consistent responses to changes in orientation. If it were placed too far from this point, tilting the front or rear of the car by the same angle could

produce different movement responses due to the difference in the sensor's position relative to the pivot point. However, a minimum tilt threshold had to be introduced to ensure that the accelerometer was not overly sensitive to small movements. While this adjustment helps prevent unintended activation, it may also create a perception of latency in the sensor response from the user's perspective.

Overall, the durability and flexibility of the LEGO structure allow the prototype to be used repeatedly and adapted for future iterations, including the integration of different sensors. This makes the design both reusable and suitable for further development and experimentation.



Fig. 2. Prototypes

4 Response to the peer critique session in Week 6

The peer critique provided valuable feedback on the design, highlighting both strengths and areas for improvement. Users found the car intuitive to handle and easy to understand, although the paper prototype was fragile. To address this, the final version was built using LEGO, providing greater durability.

Another challenge was ensuring that the buttons and sensors were easy to locate while remaining integrated into the car's design. Sensors used only during the initial recognition process were placed outside the user's direct field of view, while frequently used sensors remained visible and accessible. Additionally, a key was added to the potentiometer to make the car's activation mechanism more intuitive.

Regarding the camera interaction, users considered the candle an original idea and easily understood that it needed to be lit. However, using a lighter introduces an additional step that could make the interaction more difficult for some users.

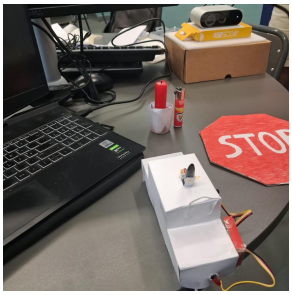


Fig. 3. Peer Critique Layout

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